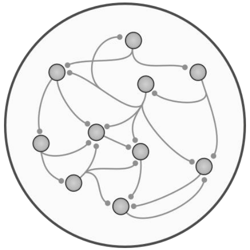
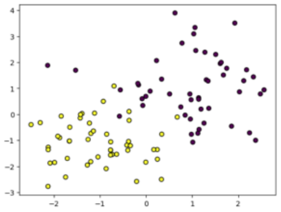
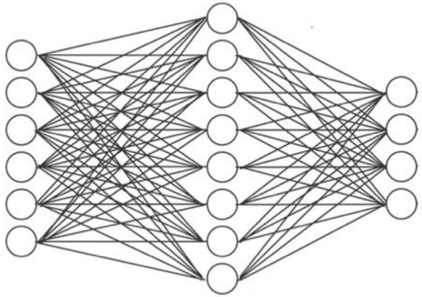
Tutorial 4: Recurrent neural networks
Master in Brain & Cognition

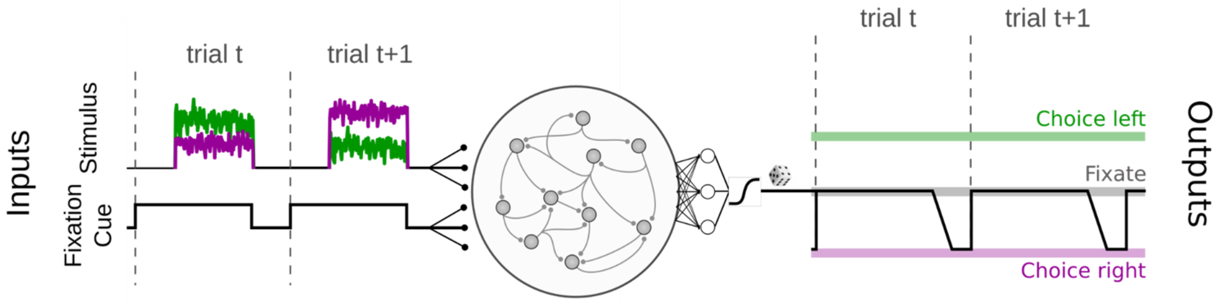


Alex Hyafil, Chris Summerfield
and Manuel Molano-Mazón (UPC)

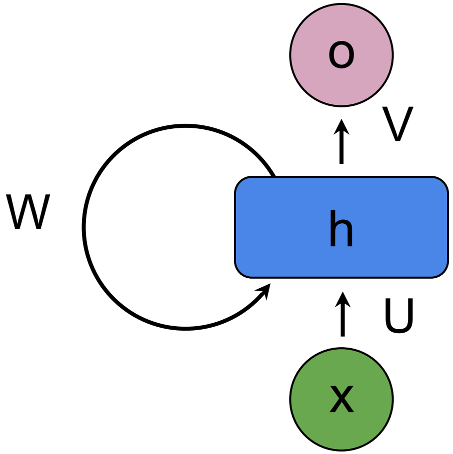
Perceptual decision-making from the point of view of a RNN



(x, y) →  → label (yelllow/blue)



Recurrent Neural Networks

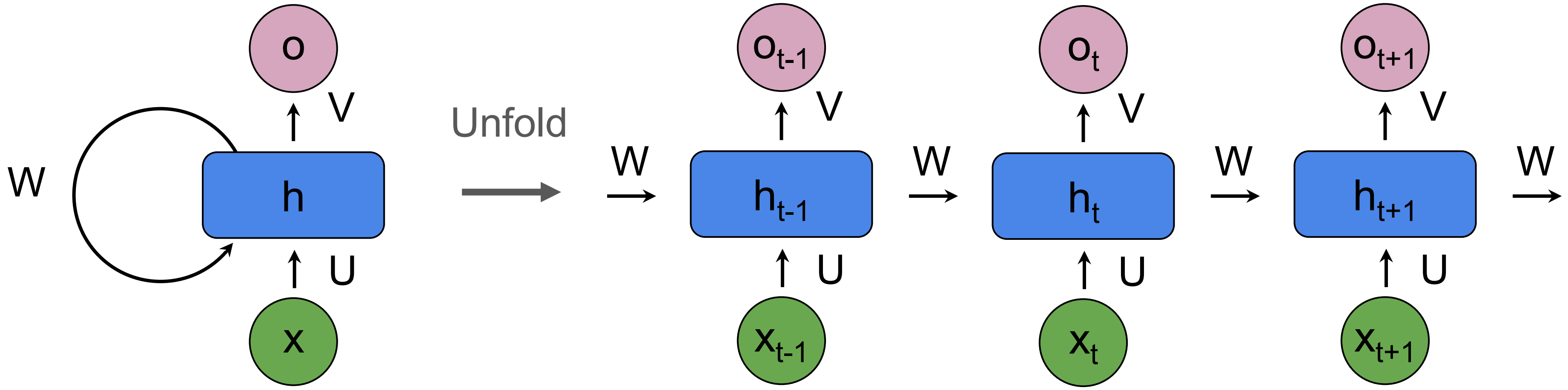


$$h_t = g\,(U \cdot X_t + W \cdot h_{t-1})$$

current input

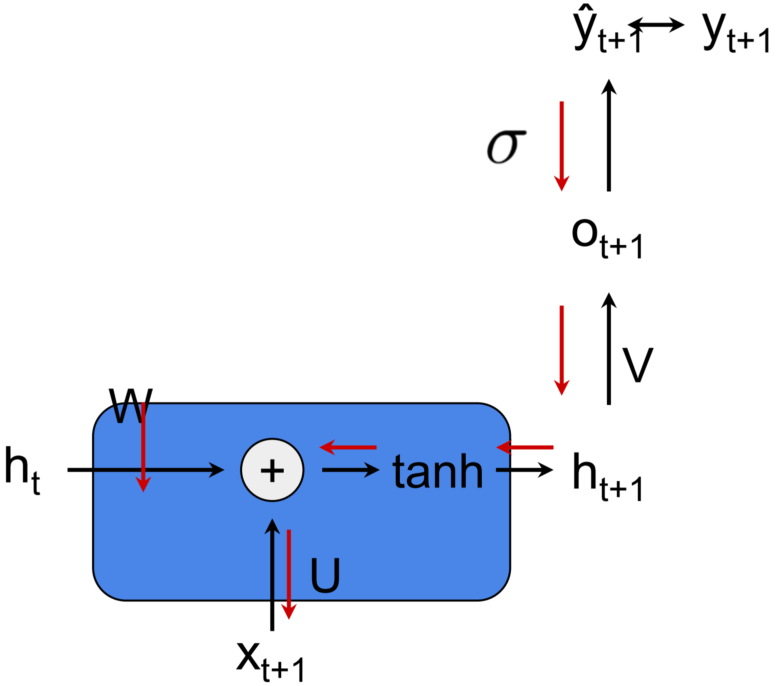
“memory”
weight matrix

previous
hidden unit
activations



o
V
W
h
U
x
Unfold
W
o_{t-1}
V
h_{t-1}
U
x_{t-1}
W
o_t
V
h_t
U
x_t
W
o_{t+1}
V
h_{t+1}
U
x_{t+1}
W

Backpropagation through time (BPTT)



$$L = -\left[\sum_{j=1}^{N} y_j \log(\hat{y}_j) + (1 - y_j)\log(1 - \hat{y}_j)\right]$$

$$a^{(t)} = \mathbf{b} + W\mathbf{h}^{(t-1)} + U\mathbf{x}^{(t)}$$

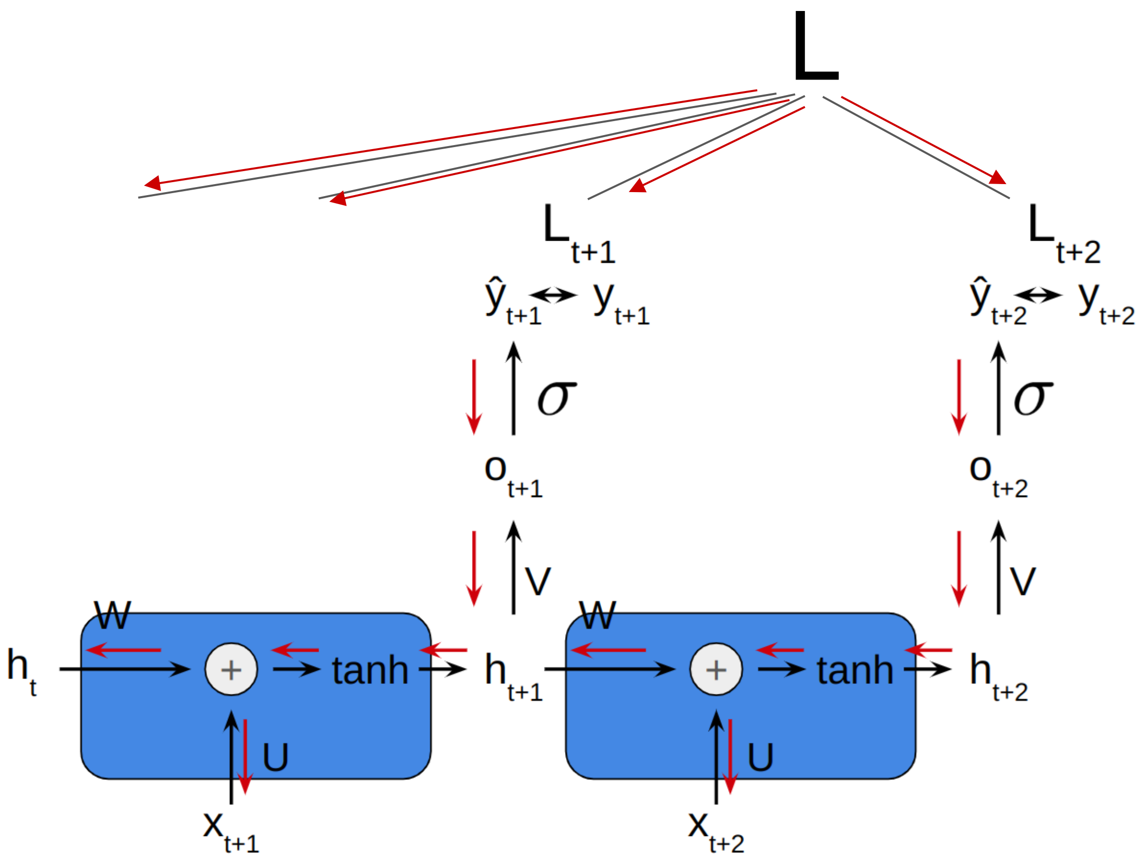
$$\mathbf{h}^{(t)} = \tanh(a^{(t)})$$

$$\mathbf{o}^{(t)} = \mathbf{c} + V\mathbf{h}^{(t)}$$

$$\hat{\mathbf{y}}^{(t)} = \sigma(\mathbf{o}^{(t)}) = \frac{1}{1 + e^{-\mathbf{o}^{(t)}}}$$

Goodfellow, Ian, et al. *Deep learning*. Vol. 1. No. 2. Cambridge: MIT press, 2016.

Backpropagation through time (BPTT)

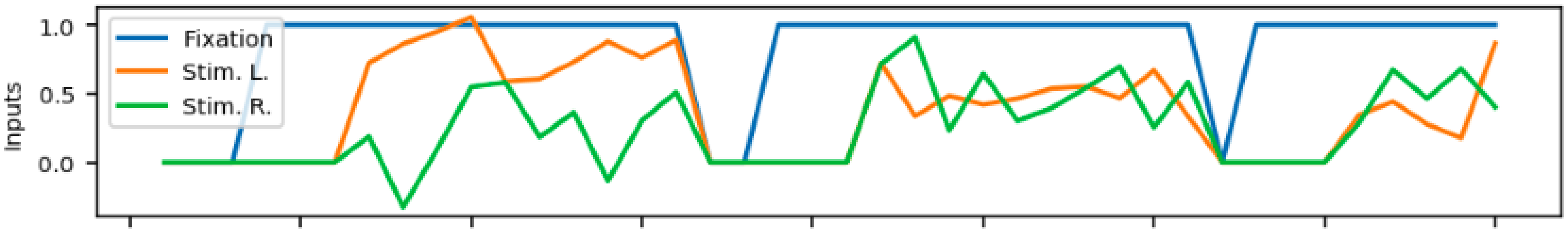


Neurogym

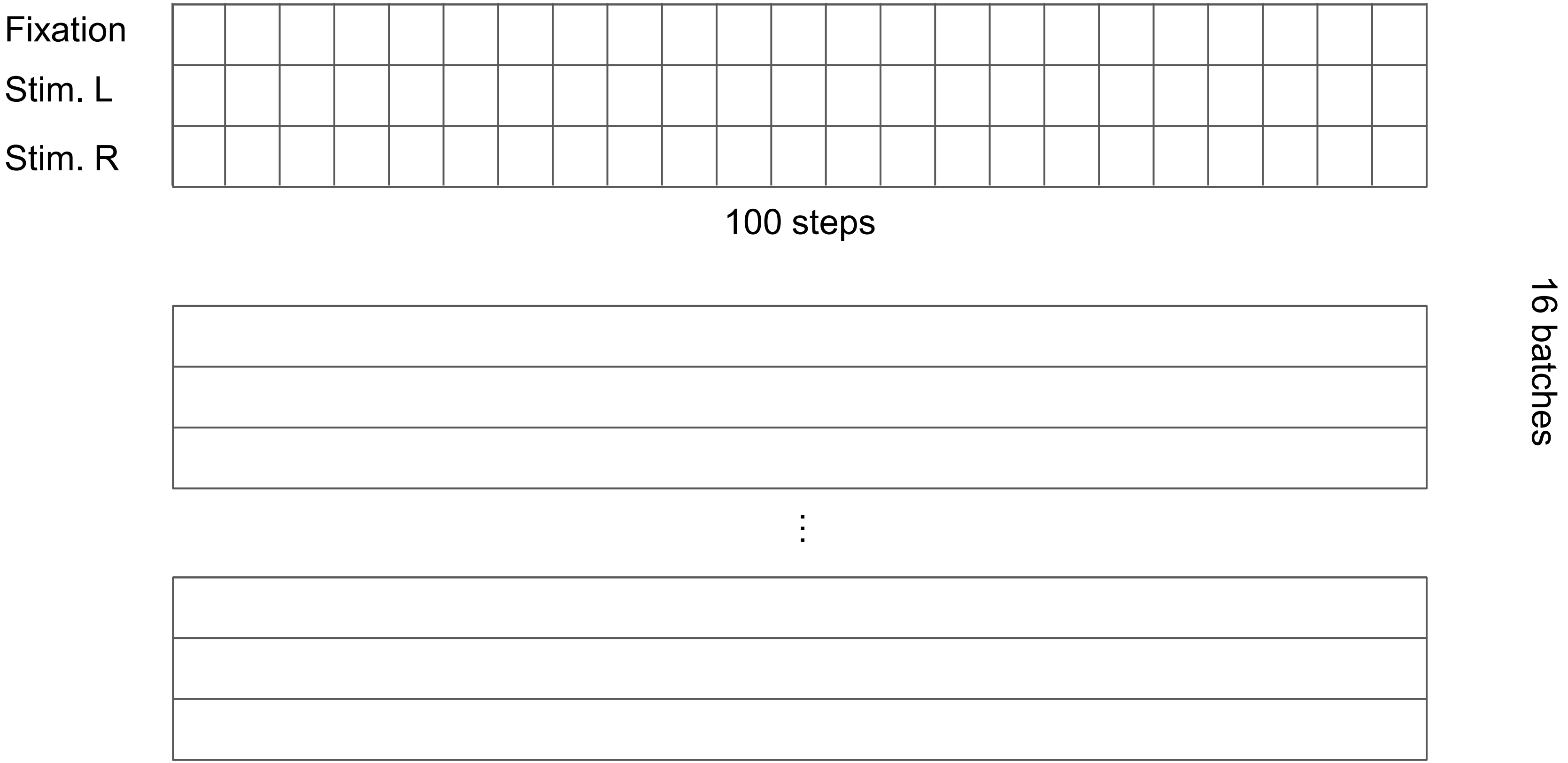
Tasks

Go/No-Go
Matching Penny
Random Dots Motion
Ready-Set-Go
Post-Decision Wager
Two-step
Delay Matching
Context-Dependent
Perceptual Decision-Making
Delay Paired Association
Multi-Bandit
Value-Based
.
.
.

Two-Alternative Forced Choice task



Regarding the inputs



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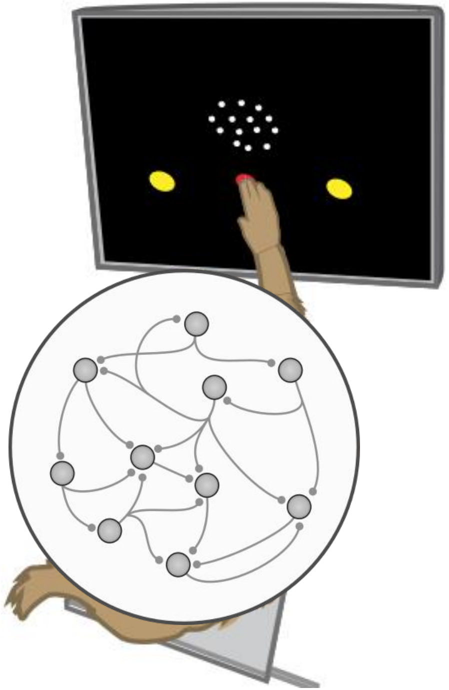
0. Installs, packages, auxiliary functions.

1. Preparing for the training:
   - Define the task (sample dataset).
   - Define the network.
   - Define the algorithm to train the network.

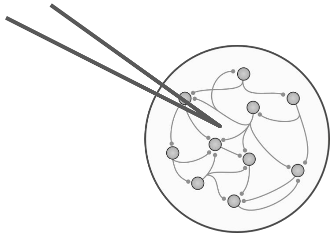
2. Supervised training of the RNN.

3. Run the trained network (and save the behavioral data).

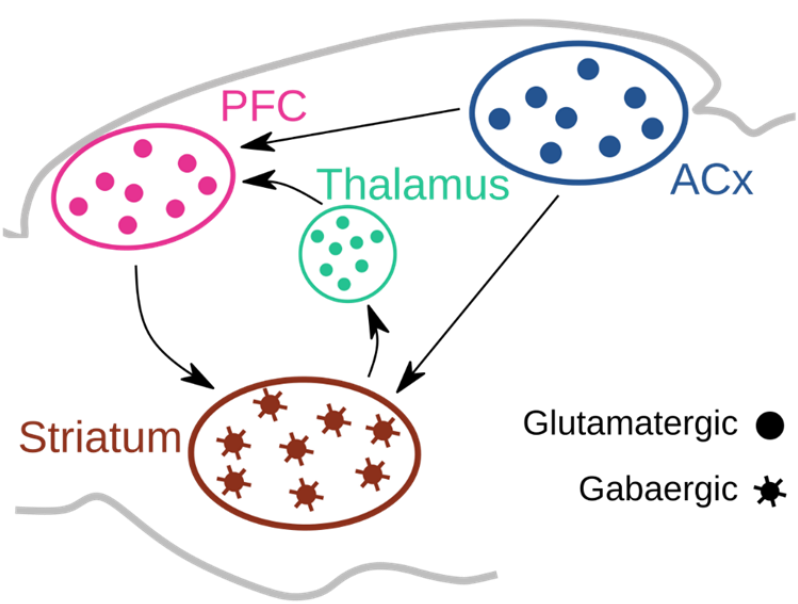
4. Network analysis:
   - Behavioral analysis.
   - General neural analysis.
   - Stimulus and choice decoding from network activity
   - Latent space analysis

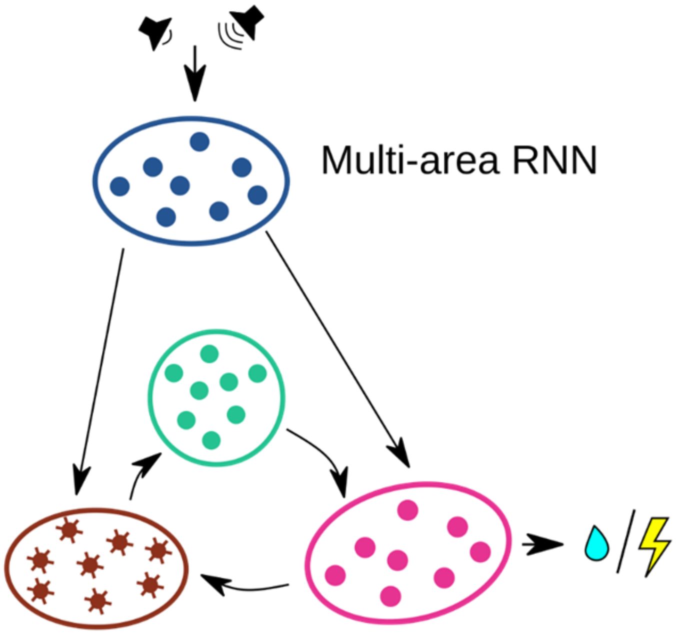


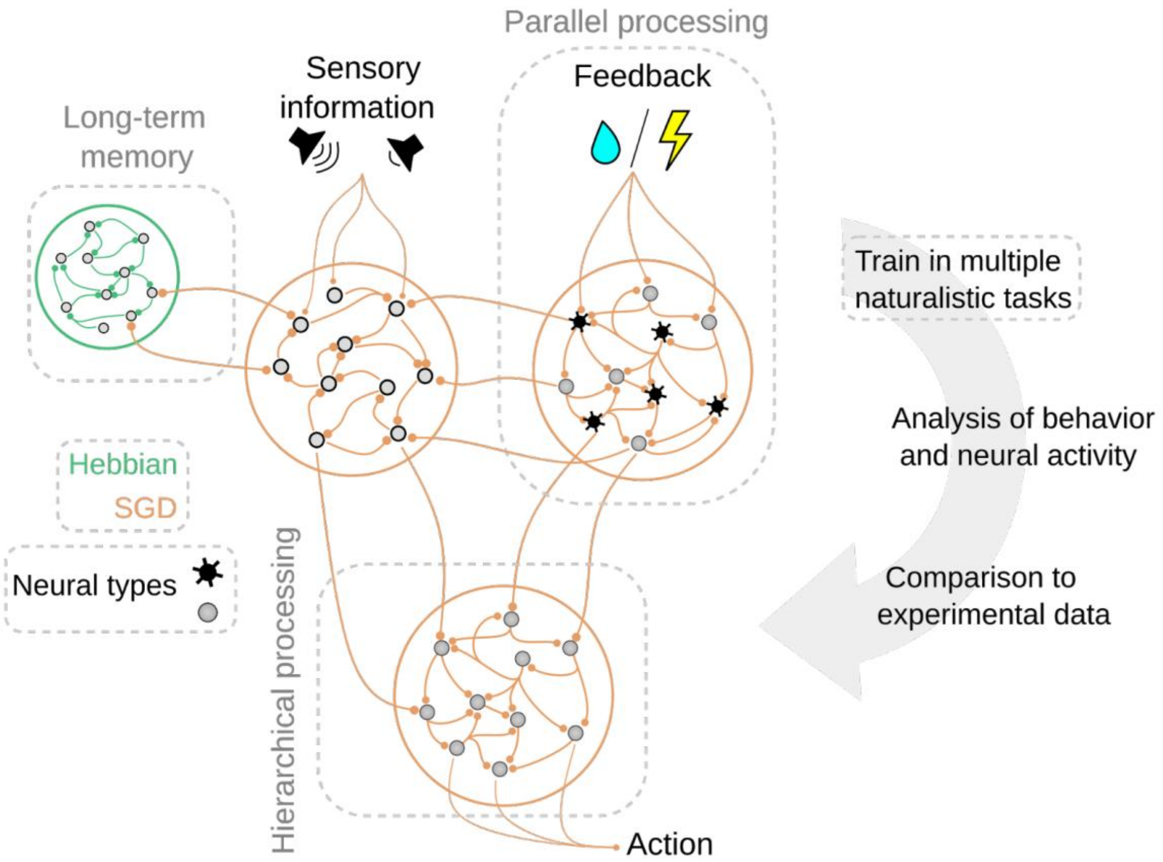
Neural analysis



Multi-area RNNs







Yang GR and Molano-Mazón M. 2021 Current Op. Neurobiol.

Yang and Wang 2020 Neuron Artificial Neural Networks for Neuroscientists: A Primer

Training an artificial neural network with Pytorch

Neurogym

Recurrent Neural Networks. Lecture 10, CS231 Course Stanford University.

Sequence Models Course, Coursera.

Andrej Karpapthy's blog. The Unreasonable Effectiveness of Recurrent Neural Networks.

Ma and Peters A neural network walks into a lab: towards using deep nets as models for human behavior

Yang and Molano-Mazón 2021 Towards the next generation of recurrent network models for cognitive neuroscience

Maheswaranathan et al. Universality and individuality in neural dynamics across large populations of recurrent networks